LABORATORY RECORD

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 1
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1. TITLE

Develop a Static Webpage Using HTML5

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2. AIM

To design and develop a static webpage using HTML5, demonstrating table-based page layout, in-page navigation links, a Home and About content area, and a Signup form with multiple input types.

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3. OBJECTIVE

The objectives of this experiment are:

- To understand the concept of a static web page as opposed to a dynamic one.
- To use an HTML table to lay out the page header, title, and navigation links.
- To structure content using the <main>, <section>, and <footer> elements.
- To create in-page navigation using anchor tags with href targets.
- To display descriptive content and a bulleted list within a two-column table layout.
- To build a Signup form using <fieldset>, <legend>, and multiple HTML input types.
- To apply basic table attributes — width, cellpadding, valign, align, and bgcolor — for layout and styling.

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4. THEORY

Static Web Page — a web page delivered to the browser exactly as stored, showing the same content to every visitor unless modified by client-side scripting such as JavaScript.

Table-Based Layout — <table>, <tr>, and <td> can be used to arrange page regions side by side, an older but still functional technique for positioning a header, navigation, and multi-column content.

Semantic Elements — <main> marks the primary content of the page, <section> groups related content such as Home or About, and <footer> marks the closing region of the page.

In-Page Navigation — an anchor tag with an href beginning with # (e.g. <a href="#home">) jumps the browser to the element on the same page whose id matches the fragment after the hash.

HTML Forms — <fieldset> and <legend> group and label a set of related form controls; input types such as text, email, and radio, together with <select>/<option>, collect different kinds of user data.

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5. ALGORITHM / PROCEDURE

- Create an HTML file named staticwebpage.html.
- Inside <head>, add the charset meta tag and the page title.
- Build a header <table> with the page title in one cell and Home / About / Signup navigation links in the other.
- Add an <hr> below the header to separate it from the main content.
- Add a <main> element containing a two-column <table> for the page's primary content.
- In the first column, add a Home <section> with a centred heading and a descriptive paragraph about static web pages.
- Below the Home section, add an About HTML <section> with a heading and an unordered list of HTML's key characteristics.
- In the second column, add a Signup <section> containing a <form> with a <fieldset> and <legend>.
- Inside the fieldset, add labelled inputs for Username, Email, and Password, radio buttons for Gender, and a Department <select> dropdown.
- Add a centred Submit button inside the form.
- Shade the Signup column with a light background colour using the bgcolor attribute.
- Add another <hr> after the main content, then a <footer> containing a full-width table with a coloured background and a centred tagline.
- Open the file in a web browser and verify the layout, navigation links, and form fields.
- Record expected vs. actual output for each test case.

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6. PROGRAM

staticwebpage.html

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
```

```
    <title> Static Weebpage </title>
</head>

<body>
    <table width="100% ">
        <tr>
            <td>
                <h1> Static Webpage </h1>
            </td>
            <td align="right">
                <a href="#home"> Home </a> | 
                <a href="#About"> About </a> | 
                <a href="#Signup"> Signup </a>
            </td>
        </tr>
    </table>
    <hr>
    <br><br>
    <main>
        <table cellpadding="30">
            <tr valign="top">
                <td width="50%">
                    <section id="#home">
                        <center>
                            <h2> Home </h2>
                            <p>
                                A static web page, sometimes called a flat page or a stationary page, is a web
page that
                                is delivered to a web browser exactly as stored, in contrast to dynamic web
pages
                                which are generated by a web application.

                                Consequently, a static web page displays the same information for all users,
from all
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                contexts, subject to modern capabilities of a web server to negotiate content-
type or
                language of the document where such versions are available and the server is
configured
                to do so. However, a webpage's JavaScript can introduce dynamic
functionality which
                may make the static web page dynamic.
              </p>
            </center>
          </section>
          <br>
          <hr>
          <section href="#About">
            <h2> About HTML </h2>
            <ul>
              <li>Simplicity: Beginner-friendly syntax using straightforward tags.</li>
              <li>Platform Independence: Runs seamlessly across any operating system.</li>
              <li>Universal Browser Support: Fully compatible with all modern
browsers.</li>
              <li>Hyperlinking: Connects internal and external web pages together.</li>
              <li>Loose Syntax: Case-insensitive tags that tolerate minor coding
mistakes.</li>
              <li>Open Standard: Free to use without any licensing fees.</li>
            </ul>
          </section>
        </td>
        <td bgcolor="#f4f4f4">
          <section id="#Projects">
          <form>
            <fieldset>
              <legend> Signup </legend>
              <p>
                <label for="Username">Enter your Username</label>
                <input type="Text" id="Username">
```

```html
                </p>
                <p>
                    <label for="Email">Enter your Email</label>
                    <input type="email" id="Email">
                </p>
                <p>
                    <label for="Password">Enter your Password</label>
                    <input type="Text" id="Password">
                </p>
                <p>
                    <label for="Gender">Gender</label>
                    <input type="radio" id="Gender" name="gender"> Male
                    <input type="radio" id="Gender" name="gender"> Female
                </p>
                <p>
                    <label for="Department"> Department </label>
                    <select id="Department">
                        <option value="CSE">CSE</option>
                        <option value="CSM">CSM</option>
                        <option value="CSD">CSD</option>
                    </select>
                </p>
                <p>
                    <center>
                        <button type="submit">Submit</button>
                    </center>

                </p>
            </fieldset>
        </form>
        </section>
      </td>
    </tr>
  </table>
</main>
```

```
    <hr>
    <br><br>
    <footer>
        <table width="100%">
            <tr>
                <td bgcolor="#91CDF8">
                    <center>
                        A Pure Semantic HTML Static Webpage
                    </center>
                </td>
            </tr>
        </table>
    </footer>
</body>

</html>
```

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7. CODE EXPLANATION

Code	Explanation
<table width="100%"> (header)	Lays out the page title and navigation links side by side.
<a href="#home">	Creates an in-page navigation link that jumps to a section on the same page.
<hr>	Draws a horizontal divider line between page regions.
<main>	Wraps the primary two-column content area of the page.
<td width="50%">	Sets a content column to half of the containing table's width.
<section>	Groups related content, such as the Home introduction or the About HTML list.
<center>	Centres the heading and paragraph text within the Home section.
/	Lists the key characteristics of HTML in the About section.
bgcolor="#f4f4f4"	Applies a light background shade to the Signup column.
<fieldset> / <legend>	Groups the Signup form fields under a labelled border.
<input type="email">	Provides a form field intended for email-formatted input.
<input type="radio" name="gender">	Restricts the Gender selection to one of the listed options.
<select> / <option>	Provides a dropdown list for selecting a Department.

Code	Explanation
<footer>	Wraps the closing table containing the page's tagline.

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8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Load the page in a browser	Header displays the title and navigation links side by side	Header rendered with title and nav links as laid out	Pass
TC-02	Click the Home navigation link	Browser jumps to the Home section	Browser navigated to the Home section	Pass
TC-03	Inspect the Home section content	Centred heading and descriptive paragraph about static web pages display	Heading and paragraph displayed centred as styled	Pass
TC-04	Inspect the About HTML section	Six list items describing HTML's characteristics render as a bulleted list	All six list items rendered correctly	Pass
TC-05	Enter values into Username, Email, and Password fields	Fields accept the typed input	All three fields accepted typed text	Pass
TC-06	Select a Gender radio button	Only one gender option remains selected at a time	Selecting one option kept only that option checked	Pass
TC-07	Select a Department from the dropdown	The chosen department option is displayed as selected	Selected department displayed correctly	Pass
TC-08	Click the Submit button	The form attempts submission using the browser's default action	Form submission was triggered	Pass
TC-09	Inspect the Signup column background	Light grey background colour displays behind the form	Background colour rendered as styled	Pass
TC-10	Inspect the footer	Tagline text displays centred with a blue background	Footer rendered with centred tagline and background colour	Pass

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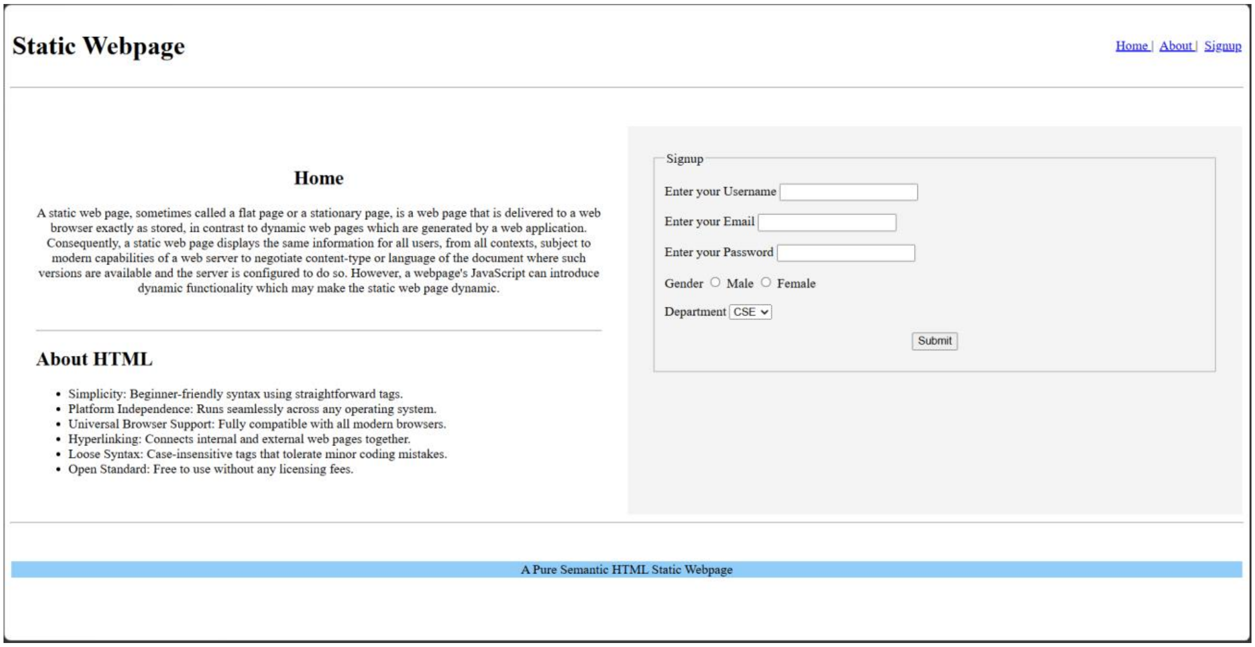
9. OUTPUT



10. RESULT

Thus, a static webpage was successfully designed and developed using HTML5. The page demonstrates table-based layout, in-page navigation, semantic content sections, and a Signup form with multiple input types. All ten test cases passed successfully, confirming correct layout, navigation, and form behaviour across the page.